

MX17: Configuration A vs B — X17 Discovery Significance

Fast-MC study with Geant4-derived detector response · 30-day exposure (578,700 pulses) · 2026-06-10

Executive summary

Config A (standard stack: MM → trigger scint → LS1 → LS2 → back-scint) reaches $Z = 3.0\sigma$ if the IPC spectrum shape is taken as perfectly known, and $Z = 2.3\sigma$ when the IPC normalisation and shape (tilt + curvature) are instead fit from the data itself. Config B (back-scint before the LS layers, used as trigger) reaches $2.3\sigma / 1.7\sigma$ on the same ladder.

B's better calorimetric coverage (42% vs 26% full calorimetry) raises its mass-channel significance by only $+0.3\sigma$, while its trigger loss (58% vs 94% of both-MM pairs) costs -0.7σ in the angle channel. With the realistic angular resolution ($\sigma_{68} \approx 14^\circ$, dominated by multiple scattering in the He-3 target walls) and mass resolution ($\sigma \approx 8$ MeV), the invariant mass adds almost no discrimination beyond the opening angle it is built from — the experiment is effectively an angular-spectrum measurement, and Config A maximises it. The $A > B$ ranking holds at every rung of the background-knowledge ladder. The X17 excess is a broad coherent surplus over $100\text{--}170^\circ$ (see stacked spectra), so the discovery power rests on how well the IPC tail shape can be constrained — by theory, sidebands, and control samples.

Results (Asimov profile-likelihood significance)

Metric	Config A	Config B
X17 pairs, angle channel (double trigger)	67	40
IPC pairs, angle channel	2110	1381
X17 pairs, mass channel (both back-scint)	18	29
IPC pairs, mass channel	1388	2022
Both-MM pair efficiency	13.0%	13.1%
Trigger efficiency (of both-MM)	94.5%	57.6%
Full calorimetry (of both-MM)	26.5%	42.2%
Z — angle, IPC shape fixed (stat only)	3.02σ	2.27σ
Z — angle, IPC normalisation fit from data	2.87σ	2.17σ
Z — angle, IPC norm+tilt+curvature fit	2.30σ	1.74σ
Z — mass, IPC shape fixed (stat only)	1.29σ	1.57σ
Z — mass, IPC norm+tilt+curvature fit	1.00σ	1.18σ

Verdict

Config A is preferred:
 $Z = 3.0\sigma$ vs 2.3σ (IPC shape known), 2.3σ vs 1.7σ (IPC shape fit from data).

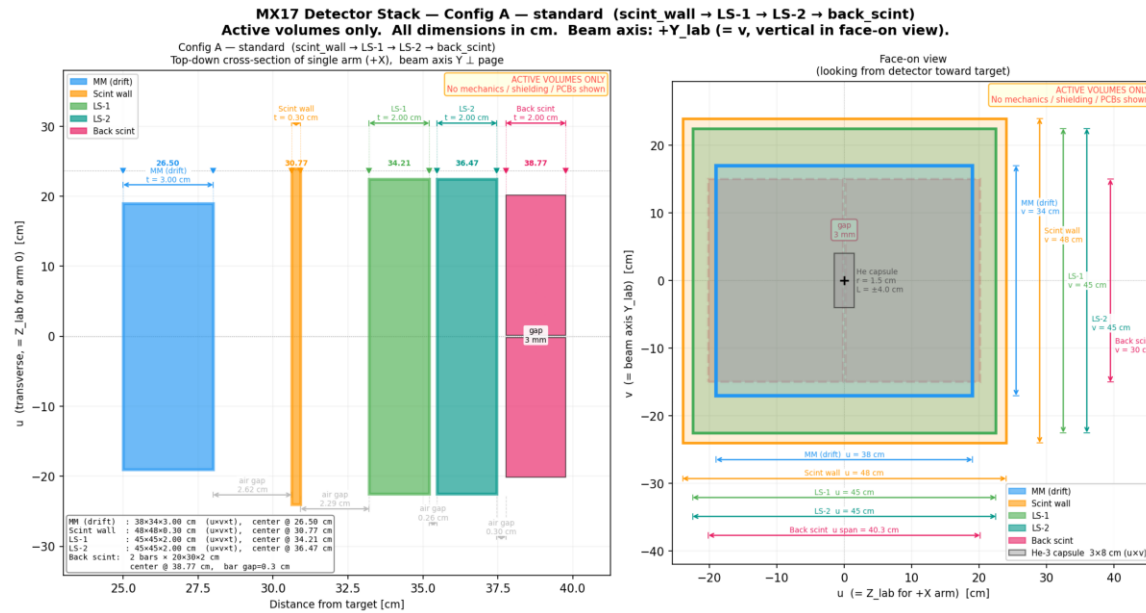
The trigger loss of Config B is not repaid by its calorimetry gain, at every level of bkg knowledge.

Caveats — read before quoting numbers

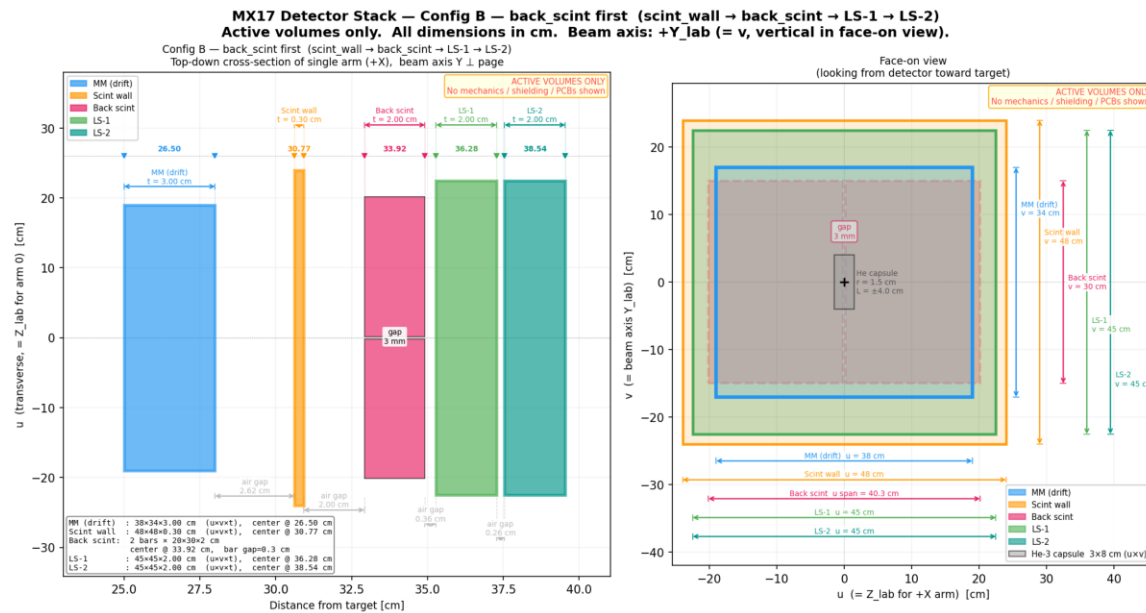
- Detector response derived from the OLD-geometry Geant4 production (22 cm arms, 300 bar He-3, 0.9 mm CFRP + 0.5 mm Al target walls, ArIso drift gas), applied to the 25 cm fast-MC layout. Regenerate the response after the new-geometry Geant4 run before quoting absolute significances.
- The X17 excess is broad (no visible bump): the stat-only Z assumes the IPC spectrum shape is exactly known. The 'fit from data' rows profile an $\exp(\text{polynomial})$ background over the full range — the truth lies between, depending on theory/sideband constraints on the IPC tail.
- Config B's calorimeter energy response reuses the standard-stack (Config A) tables — adequate for this relative comparison only.
- Rates: IPC = $1.12\text{e-}2/\text{pulse} \div 0.3$ efficiency scale (Alberto), X17/IPC = 2.5%. Z scales as $\sqrt{(\text{exposure})}$ and linearly with the X17 fraction. Timing: 30 ns resolution, 200 ns coincidence window (lossless).
- No cosmic/beam-related backgrounds beyond IPC.

Detector stack geometry (top-down cross-section + face-on view)

Config A — standard: MM → trigger scint → LS-1 → LS-2 → back-scint



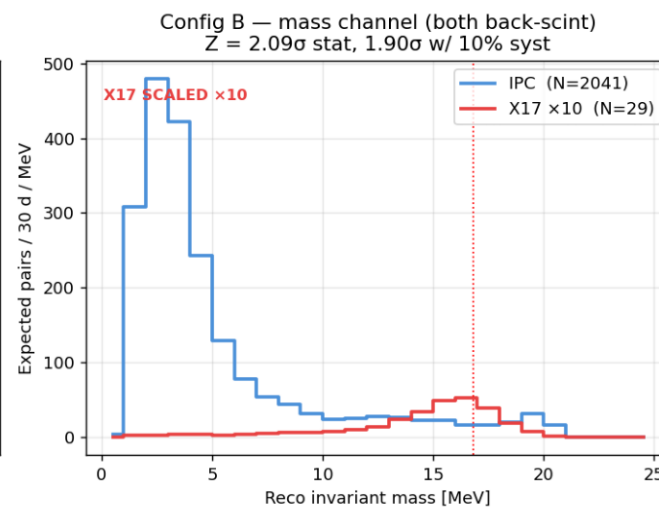
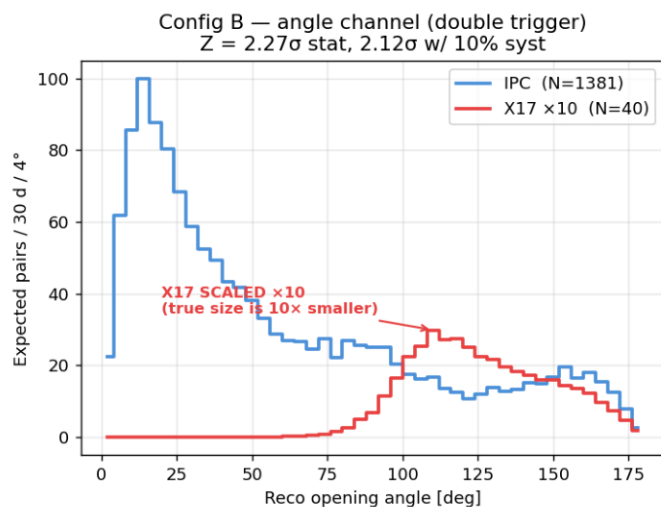
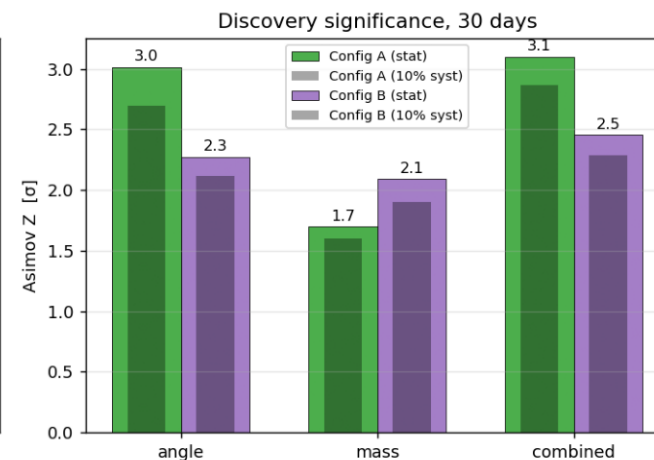
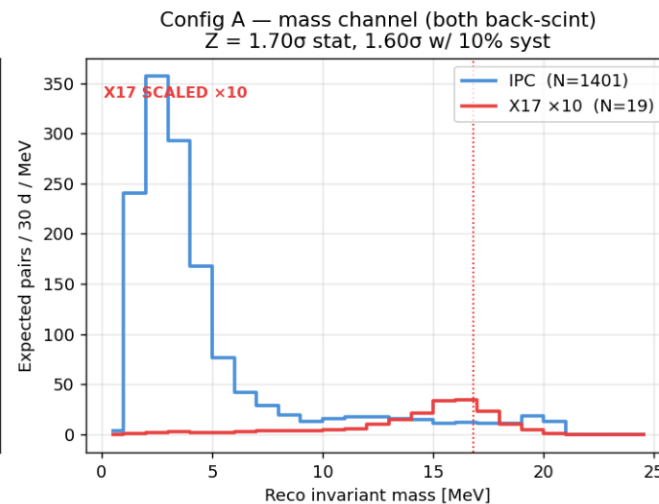
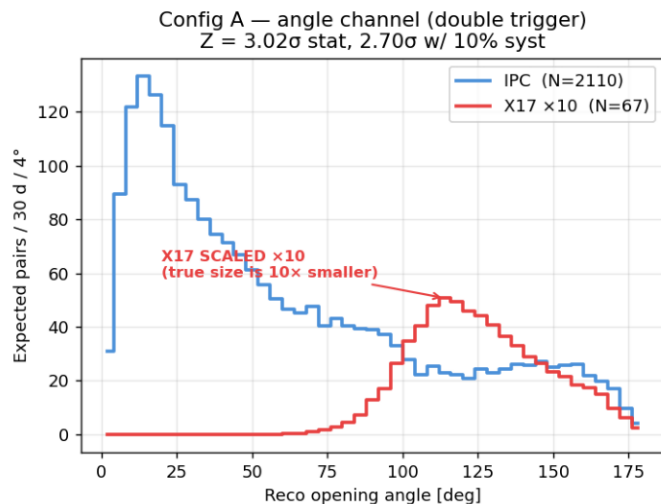
Config B — back-scint first: MM → trigger scint → back-scint → LS-1 → LS-2



Active volumes only; dimensions in cm; beam along +Y.

Expected spectra and significance — 30-day exposure

X17 vs IPC discovery significance — Configuration A vs B



Exposure: 30 d, 578,700 pulses
Produced: X17 540, IPC 21605
MC: 200,000 windows / source / config

Geant4 response: direction MS smearing +
calorimetric mass (old-geometry tables)

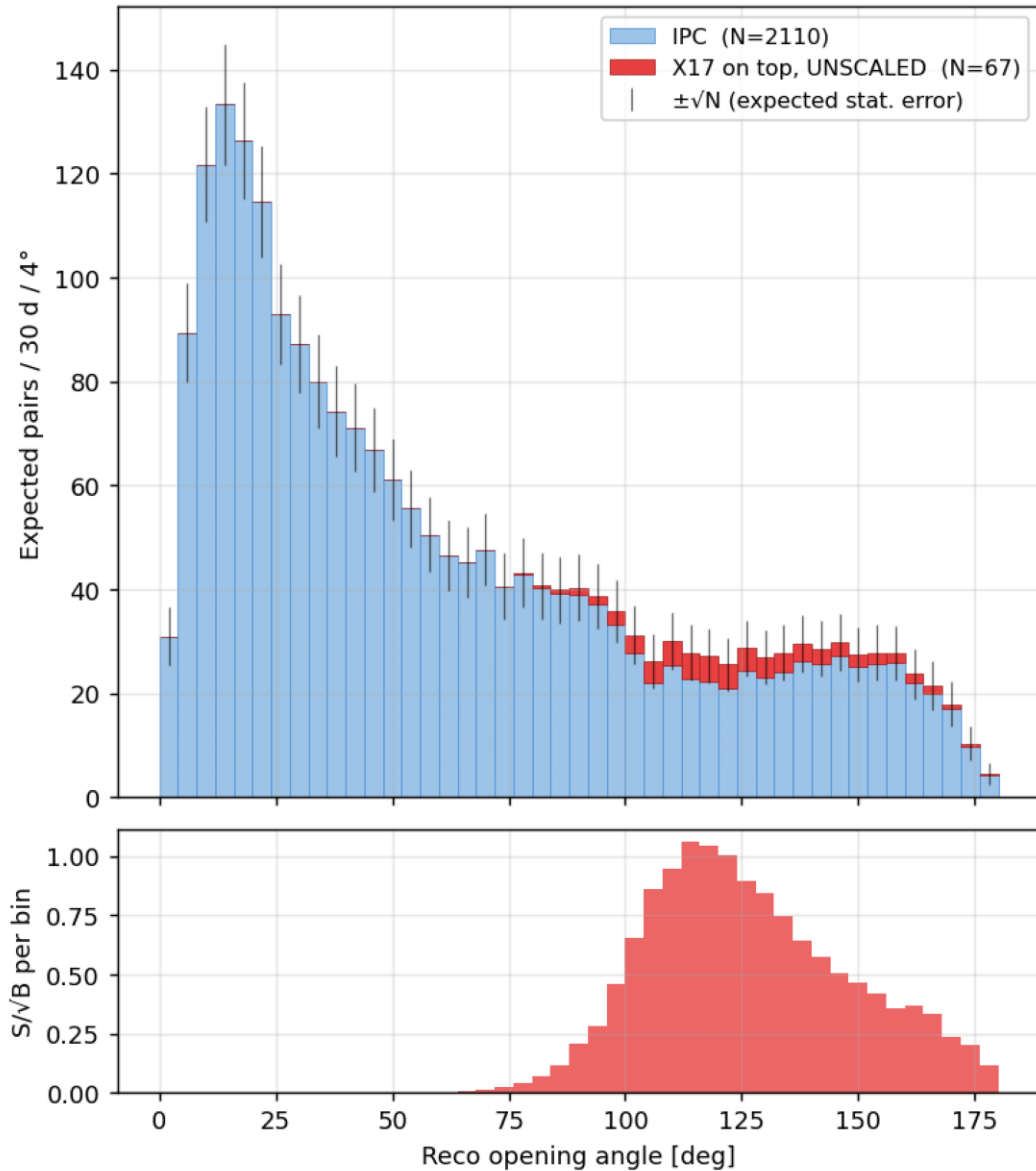
Config A: both-MM eff 13.1%, trig eff 94.4%, calor. compl. 26.5%
Config B: both-MM eff 13.0%, trig eff 56.5%, calor. compl. 41.9%

Unscaled stacked spectra — the X17 excess as it appears in data

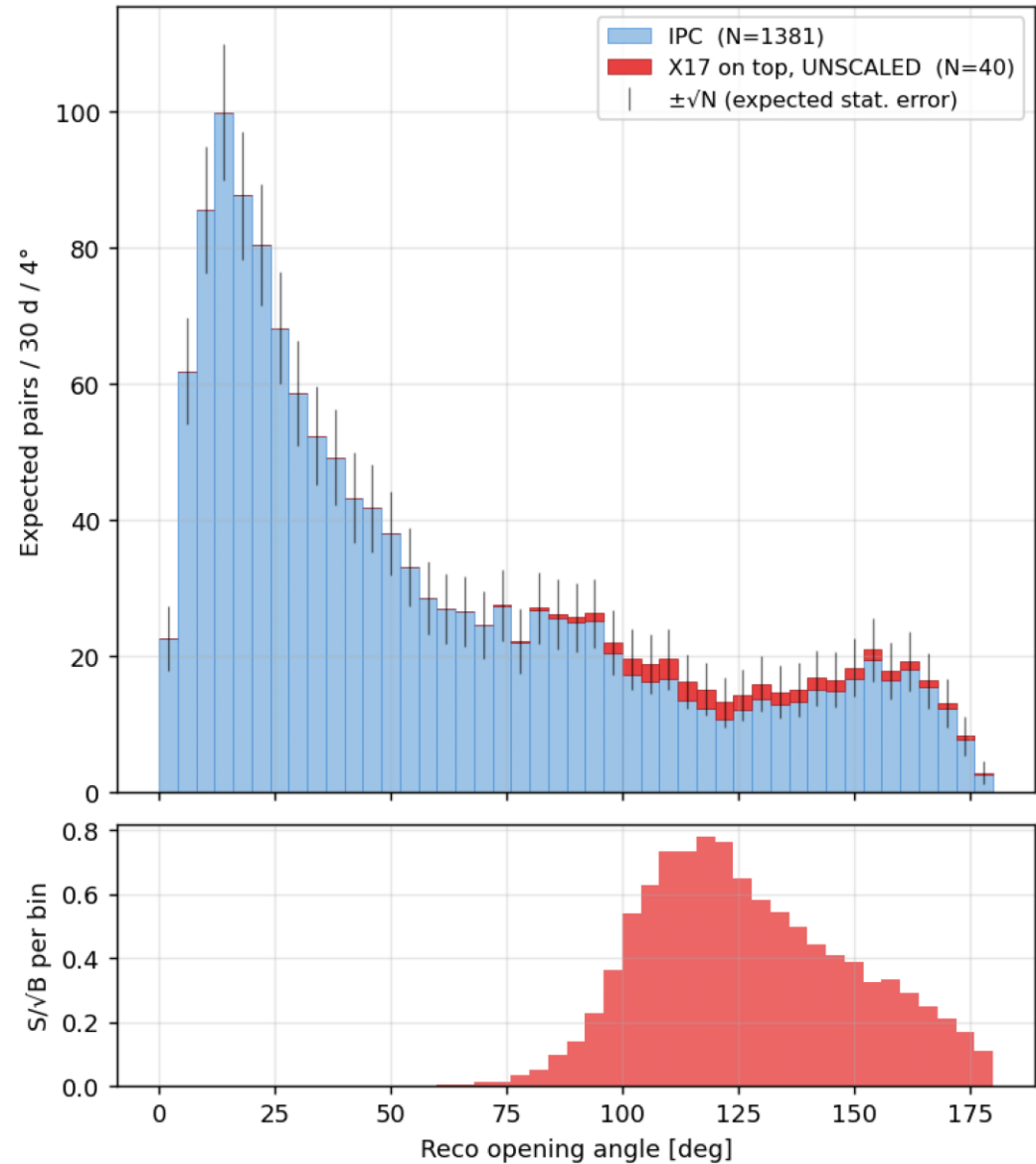
Unscaled stacked spectra — the X17 excess as it will appear in data

Significance comes from the COHERENT excess across many bins (per-bin $S/\sqrt{B}$ below), not from any single visible bump

Config A — angle channel, stacked & unscaled
(the X17 sliver is the entire signal)



Config B — angle channel, stacked & unscaled
(the X17 sliver is the entire signal)



Enhancement scan — symmetric-energy selection (no significant gain)

Enhancement scan: symmetric-energy selection (Geant4-response toy, MM-double acceptance)

